

Handwritten Form Recognition using YOLO and TrOCR with Layout Aware Error Correction

1. Team Details

- **Team Name:** Tokenwise
- **Team Leader Name:** Satishkumar Thirumalai
- **Team Leader Institution and Email:**
Indira Gandhi Centre for Atomic Research, sathishkumartheta@gmail.com
- **Rest of the Team Members:**
Anand Rajendran, Freelancing Consultant
Muthukumar G, Indira Gandhi Centre for Atomic Research
Iniya Sargam,
- **Team Website URL (if any):** NA
- **Affiliations:**
Satishkumar Thirumalai¹
Anand Rajendran¹
Muthukumar G¹
Iniya Sargam¹
- **Link to Codes/Executables:** Package Delivered
- **Link to Enhancement Results:** Package Delivered

2. Contribution Details

- **Title of the Contribution:**
Handwritten Form Recognition using YOLO and TrOCR with Layout Aware Error Correction
- **General Method Description:**
We present a robust handwritten form recognition pipeline that effectively leverages both spatial and semantic cues to achieve accurate text extraction from scanned form documents.

The first stage of the pipeline involves handwritten field detection using a fine-tuned YOLOv8 object detection model. This model is trained to localize regions containing handwritten text within form images. It outputs separate bounding box for each predicted field region.

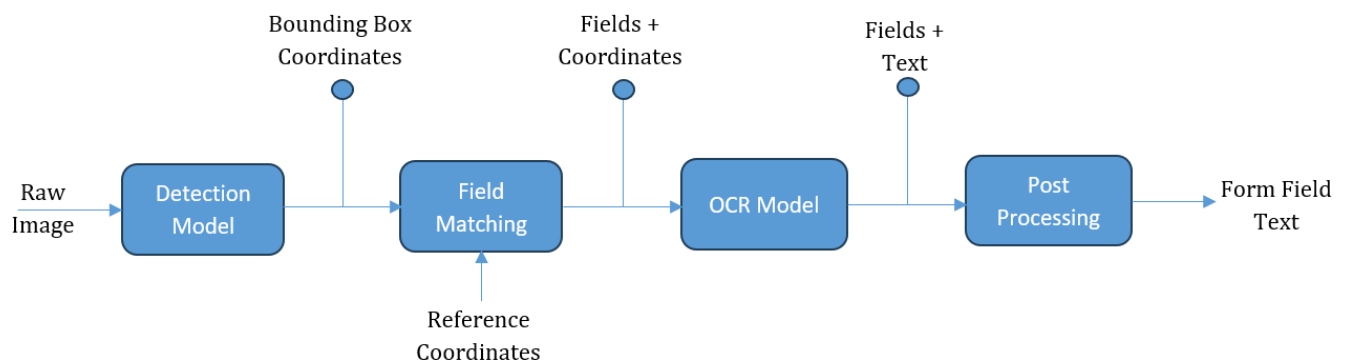
Following detection, a field matching algorithm is applied. Each predicted bounding box is compared with a set of reference ground-truth boxes obtained from template forms. Using Intersection over Union (IoU) as a matching criterion, the system assigns each detected field to a corresponding predefined field name. This step is critical for contextual understanding and for applying field-specific corrections later in the pipeline.

Once the fields are matched, the cropped image regions are passed to a pre trained TrOCR model on the handwritten dataset which performs the actual handwritten text recognition. TrOCR converts the handwritten content in each cropped region into a raw textual prediction using a transformer architecture.

To improve the reliability of the recognized text, we proposed an IoU guided text post-processing stage. Based on the semantic label assigned to each field (e.g., gender, date, or phone number), additional correction logic is applied. For example, if the field is a phone number, spurious alphabetic characters are removed to retain only digits. Similarly, for categorical fields like gender, 1/2 character prediction errors such as "Malee" is corrected to "Male" using learned patterns.

The final output is a structured and cleaned representation of the form, serialized in formats such as XML-like tags or JSON. This enables downstream systems to reliably parse and store extracted information for further processing, analytics, or integration into digital workflows.

- **Representative Image:**



- **Loss Function:**

- For YOLO, Box Loss
- For TrOCR, Cross Entropy Loss

- **Testing of Previously Published Methods:**

Model	Task	Results
YOLOv8	Handwritten Field Detection	Bounding box prediction Precision = 98.8 Recall = 98.61 map@50 = 78.58
TrOCR	Handwritten Text Recognition	CER = 29.79, WER = 114.12
Tesseract	Handwritten Text Recognition	CER = 72.54, WER = 159.32

- **Use of Extra Data:** No

- **Other Methods and Baselines Tested:**

Custom OCR CTC Model: The OCR model is a convolutional recurrent neural network trained with Connectionist Temporal Classification (CTC) loss to recognize handwritten text from grayscale images. The model architecture begins with a convolutional backbone consisting of multiple Conv2D layers with increasing filter sizes (32, 64, 128), each followed by max pooling and batch normalization to extract spatial features. The final convolutional output is reshaped into a sequence format suitable for temporal modeling. This sequence is then processed by two stacked bidirectional LSTM layers with 128 hidden units each, enabling the model to capture contextual dependencies in both forward and backward directions. A dense layer with softmax activation generates probability distributions over the character vocabulary for each time step. For training, a custom CTC loss layer computes the alignment-free sequence loss between predicted character probabilities and ground truth labels. The architecture supports two modes: a training model that incorporates the CTC loss, and a separate prediction model that outputs character probabilities for decoding during inference. This setup allows the model to handle varying text lengths and unsegmented input efficiently.

3. Global Method Description

- **Total Method Complexity (all stages):** Moderate

The pipeline integrates object detection (YOLO), spatial matching (IoU), text recognition (TrOCR), and post-processing corrections. Inference and post-processing are lightweight after initial training

- **Pre-trained or External Models Used:**

- YOLOv8: Pretrained on COCO, fine-tuned on handwritten form bounding boxes
- TrOCR Large (Handwritten): Pretrained on IAM dataset

- **Additional Data Used:** No
- **Training Description:**
 - Framework: PyTorch + Hugging Face Transformers + Ultralytics YOLOv8
 - Hardware: NVIDIA GPU P100 on Kaggle
 - Input: Scanned Images of forms
 - Ground Truth: Bounding boxes and corresponding structured text annotations
 - Training Data: 1050 Forms with 80/20 split
 - Augmentation: No heavy augmentations are used
 - Optimizer: SGD with momentum (YOLO), AdamW (TrOCR)
 - Learning Rate: 0.01 (YOLO),
 - Epochs: 50
- **Testing Description:**
 - Form images to YOLO for detection
 - Detected boxes matched with reference fields via IoU
 - Cropped regions resized and normalized
 - OCR performed using TrOCR on the cropped image
 - Text post-processed based on field labels
- **Quantitative and Qualitative Advantages:**
 - Parameter count:
 - YOLOv8: 11M parameters
 - TrOCR: 570M parameters
- **Results of Comparison to Other Approaches:** Shown in the Table below
- **Results on Other Benchmarks:** NA
- **Novelty and Prior Publication:**
 - Novel combination of YOLOv8 + IoU-based label alignment + TrOCR + post-processing
 - No prior work observed with this specific layout-based correction loop
 - Not previously published in any conference or journal

4. Technical Table

The performance of the proposed and baseline models on the test set are summarized in the table below.

Model	CER	WER	Field Accuracy	Final Score
YOLO + Tesseract	72.54	159.32	2.24	-10.81
YOLO + Custom OCR	37.58	46.93	62.58	40.51
YOLO + TrOCR	29.79	114.12	6.27	20.56
YOLO + TrOCR + Layout Aware Post Processing	15.82	41.62	47.52	57.02

5. Competition Particularities

- **Any particularities of the solution for this competition compared to other challenges:**

The standard evaluation metrics such as Character Error Rate (CER), which measures the percentage of incorrect characters, Word Error Rate (WER), which accounts for insertions, deletions, and substitutions at the word level, Field Accuracy, which represents the proportion of fields correctly recognized in their entirety and lastly the final score is the weighted sum of CER, WER and Field Accuracy is used in this challenge.

6. References:

- [1] Minghao Li¹, Tengchao Lv², Jingye Chen², Lei Cui², Yijuan Lu², Dinei Florencio², Cha Zhang², Zhoujun Li¹, Furu Wei²: "TrOCR: Transformer-based Optical Character Recognition with Pre-trained Models", published in arXiv, 2022
- [2] Nanyang Yang, Yi Wang, Lap-Pui Chau: "Multi-object Tracking with Tracked Object Bounding Box Association", published in ICME workshop, 2021
- [3] Yongsun Yoon: "Fine-tuning YOLOv8 with Custom Dataset Generated by Open-world Object Detector", published in Medium, [Link](#)

- [4] Neeraj Garg, Neelam Sharma, Gautam Jain, Vipul Jain, Vaibhav Upreti: "Handwriting Recognition System Using YOLO and CTC", published in International Conference on Advanced Computing & Communication Technologies (ICACCTech), 2023
- [5] Gang. Li, Zhao. Wang, Junhui. Huang, Jianmin. Gao, Kun. Chen: "Form Recognition Based on Lightweight U-Net and Tesseract after Multi-level Retraining", published in International Conference of Optical Imaging and Measurement (ICOIM), 2021
- [6] A Nikitha, J Geetha, D.S JayaLakshmi: "Handwritten Text Recognition using Deep Learning", published in International Conference on Recent Trends on Electronics, Information, Communication & Technology (RTEICT), 2020
- [7] Juan Ramirez-Orta, Eduardo Xamena, Ana Maguitman, Evangelos Milios, Axel J. Soto: "Post-OCR Document Correction with large Ensembles of Character Sequence-to-Sequence Models", published in arXiv, 2022
- [8] Quoc-Dung Nguyen, Duc-Anh Le, Nguyet-Minh Phan, Ivan Zelinka: "OCR error correction using correction patterns and self-organizing migrating algorithm", published in Springer Nature Link, 2021